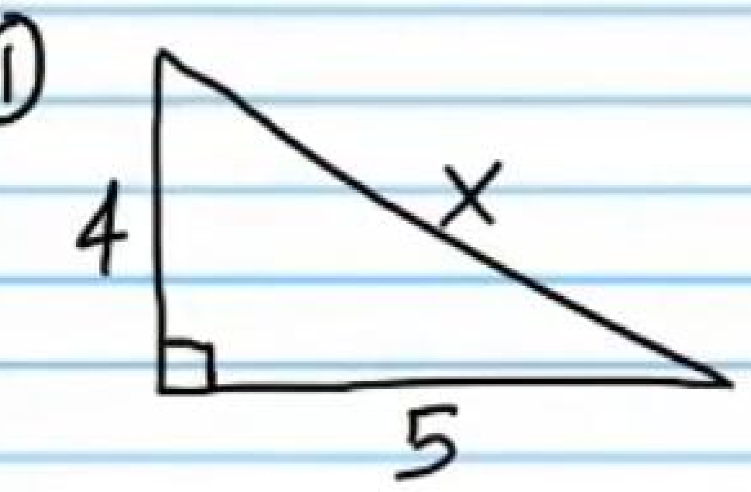
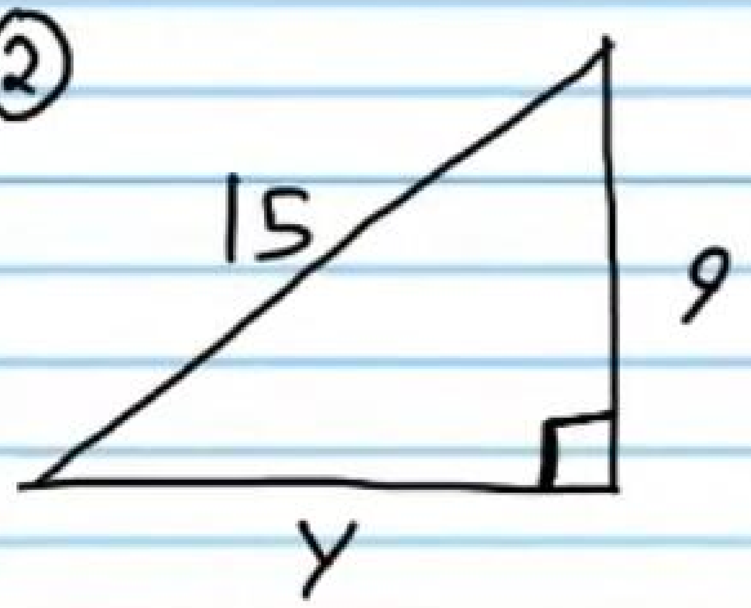
Name
Date
Course

Right Triangles (Part I): Geometric Mean Theorems, Special Right Triangles, and Pythagorean Theorems (Homework).

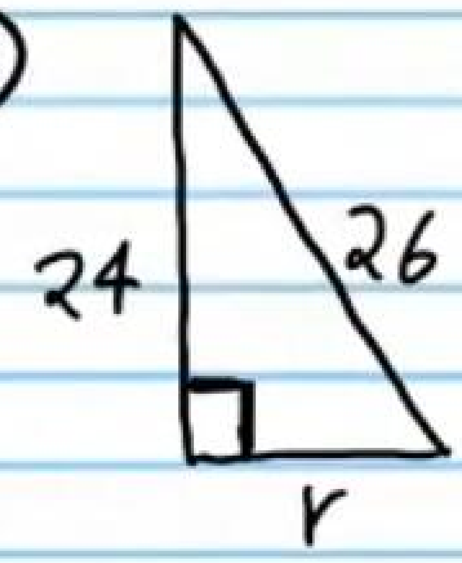
Find the missing side of each right triangle.

1


$\boxed{x = \sqrt{41}}$

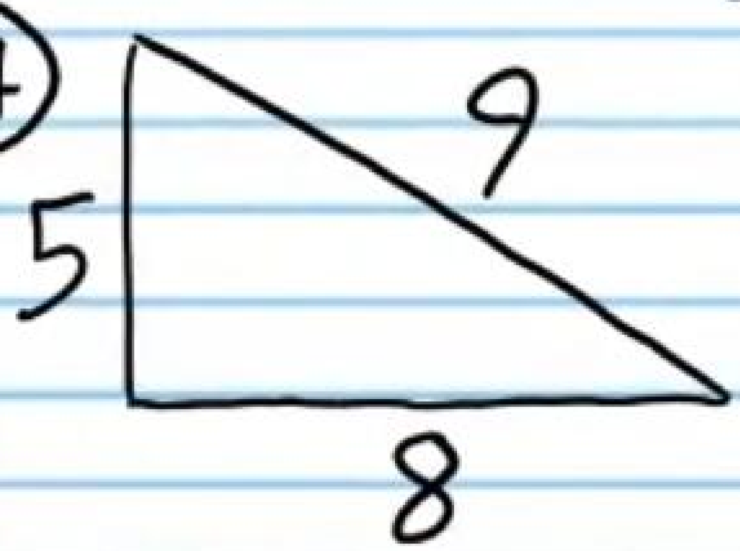
2


$\boxed{y = 12}$

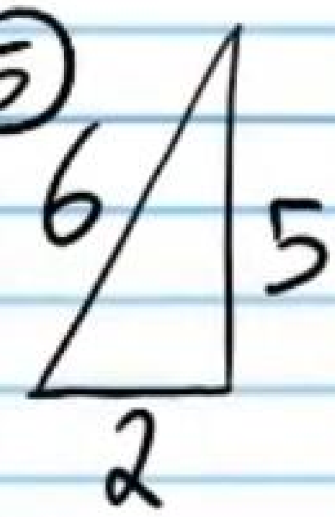
3


$\boxed{r = 10}$

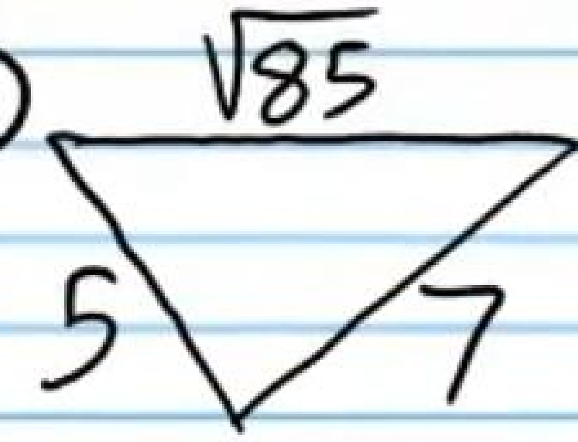
Are the following triangles acute, obtuse, or right triangles?

4


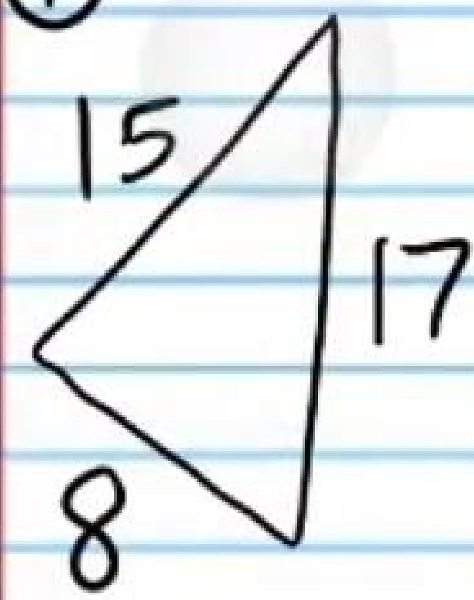
Acute

5


Obtuse

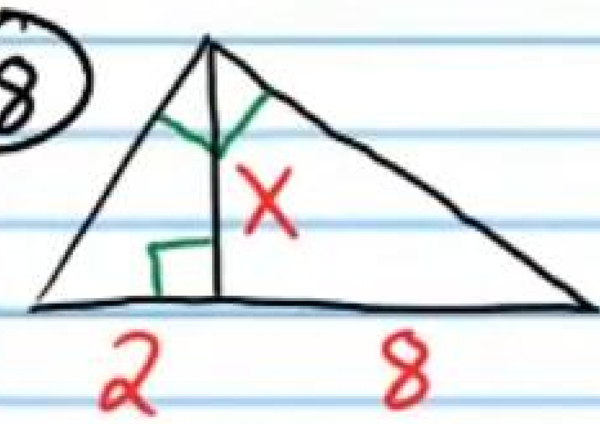
6


Obtuse

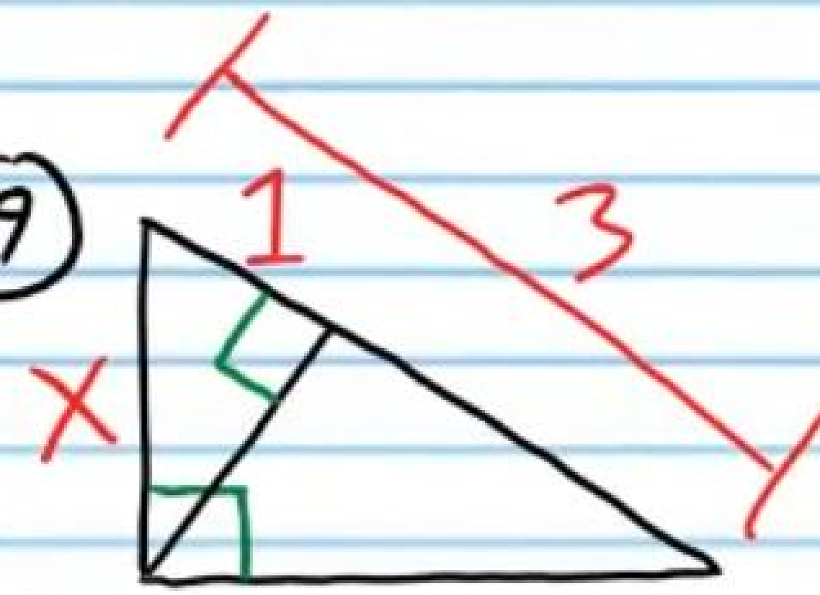
7


Right

Find the unknown values.

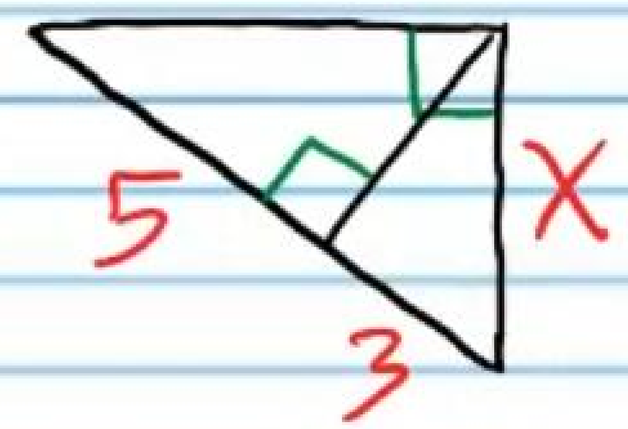
8


$\boxed{x = 4}$

9


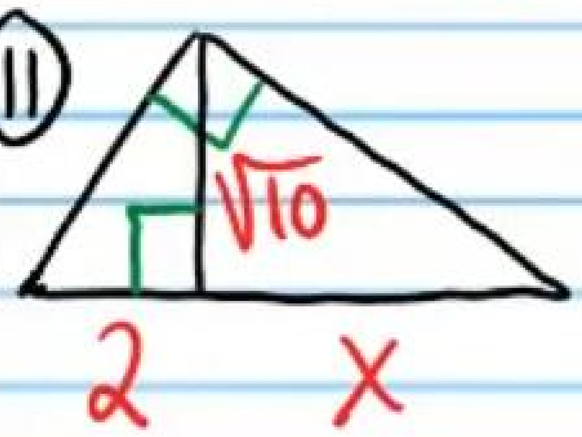
$\boxed{x = \sqrt{3}}$

(10)



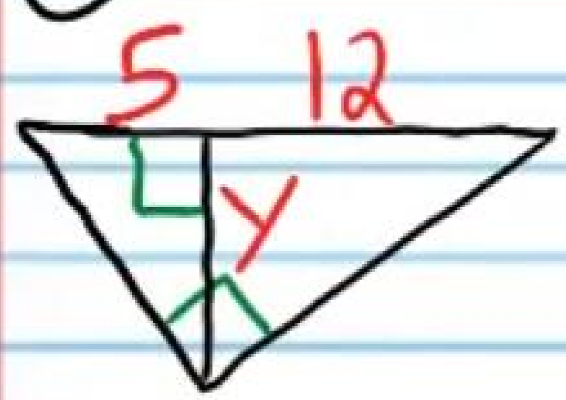
$\boxed{x=2\sqrt{6}}$

(11)



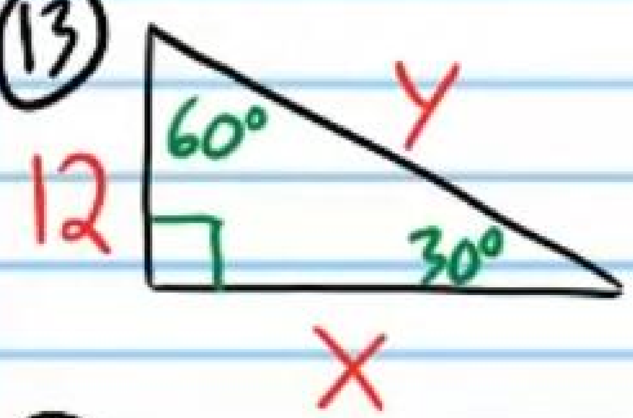
$\boxed{x=5}$

(12)



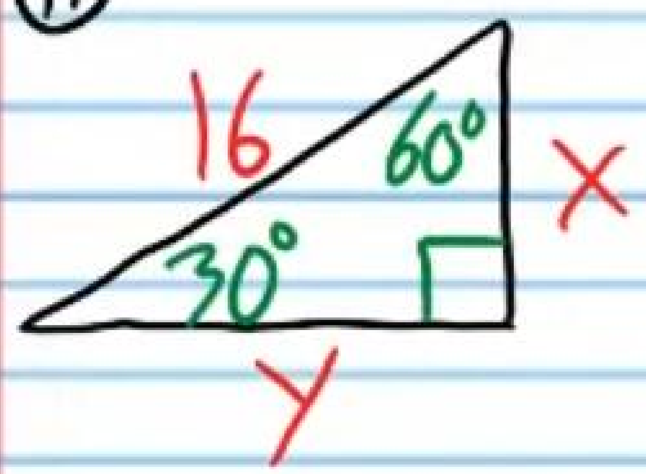
$\boxed{y=2\sqrt{15}}$

(13)



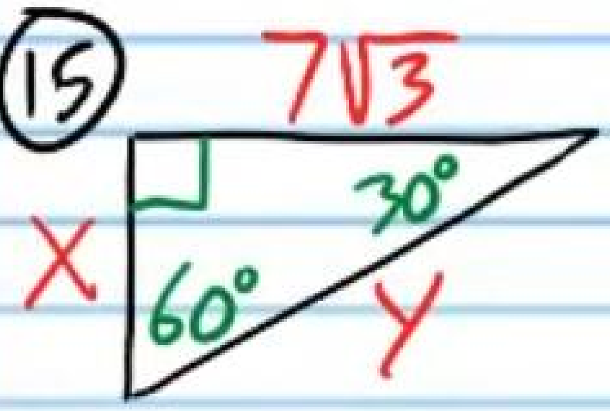
$\boxed{\begin{array}{l} x=12\sqrt{3} \\ y=24 \end{array}}$

(14)



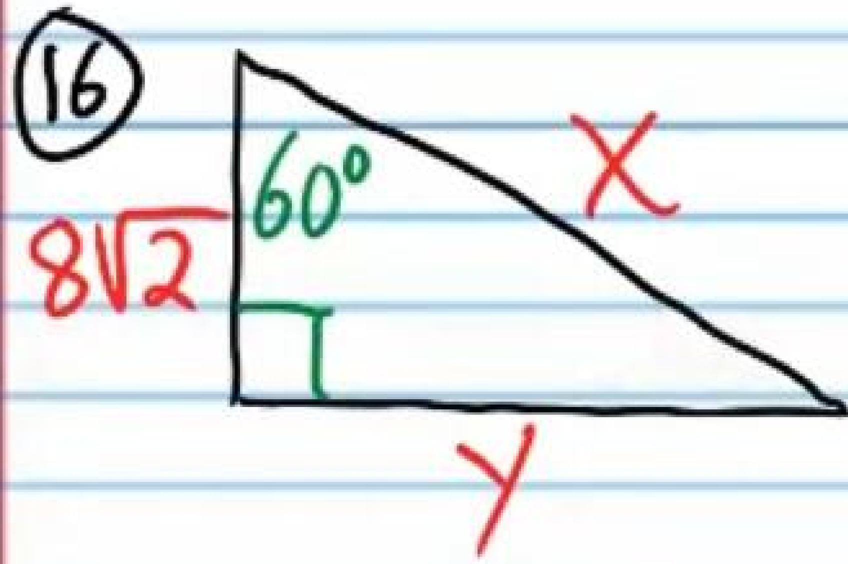
$\boxed{\begin{array}{l} x=8 \\ y=8\sqrt{3} \end{array}}$

(15)



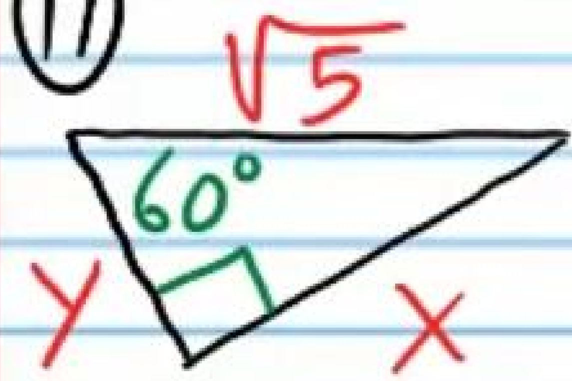
$\boxed{\begin{array}{l} x=7 \\ y=14 \end{array}}$

(16)



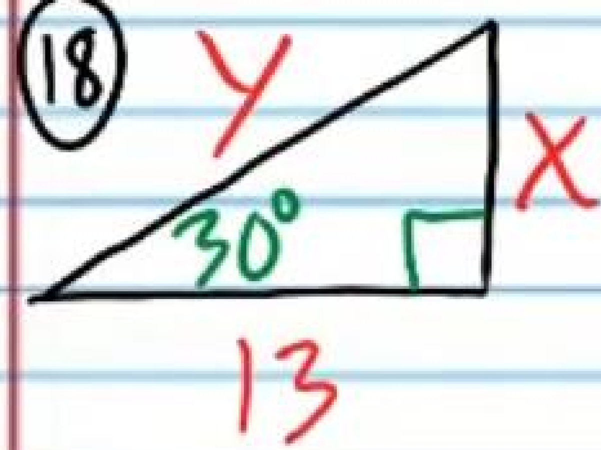
$\boxed{\begin{array}{l} x=16\sqrt{2} \\ y=8\sqrt{6} \end{array}}$

(17)



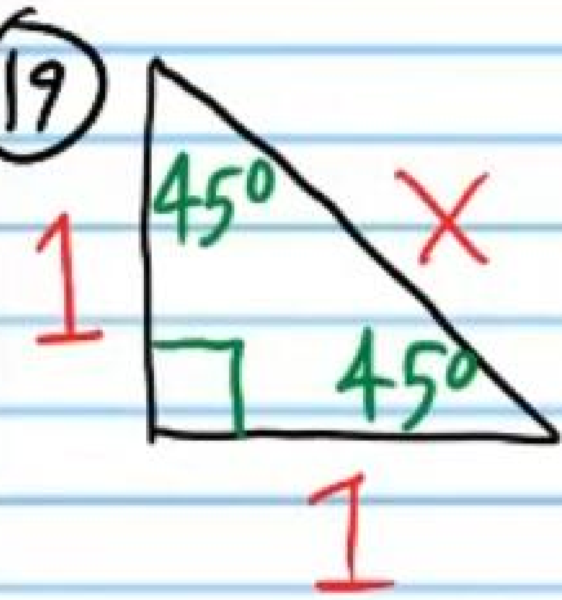
$\boxed{\begin{array}{l} y=\frac{\sqrt{5}}{2} \\ x=\frac{\sqrt{15}}{2} \end{array}}$

(18)



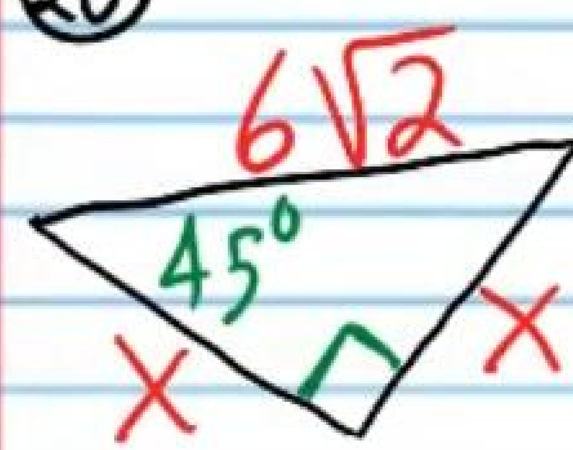
$\boxed{\begin{array}{l} x=\frac{13\sqrt{3}}{3} \\ y=\frac{26\sqrt{3}}{3} \end{array}}$

(19)



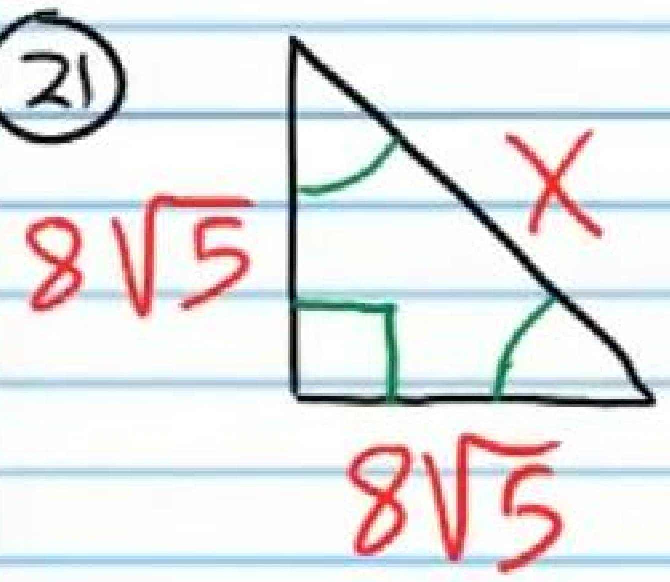
$\boxed{x=\sqrt{2}}$

(20)



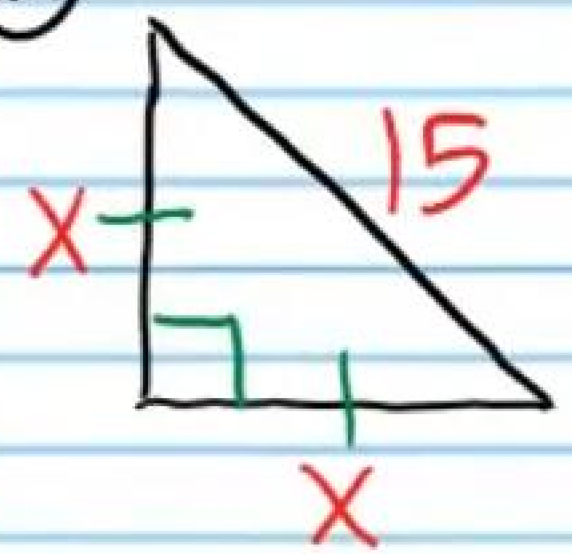
$\boxed{x=6}$

21



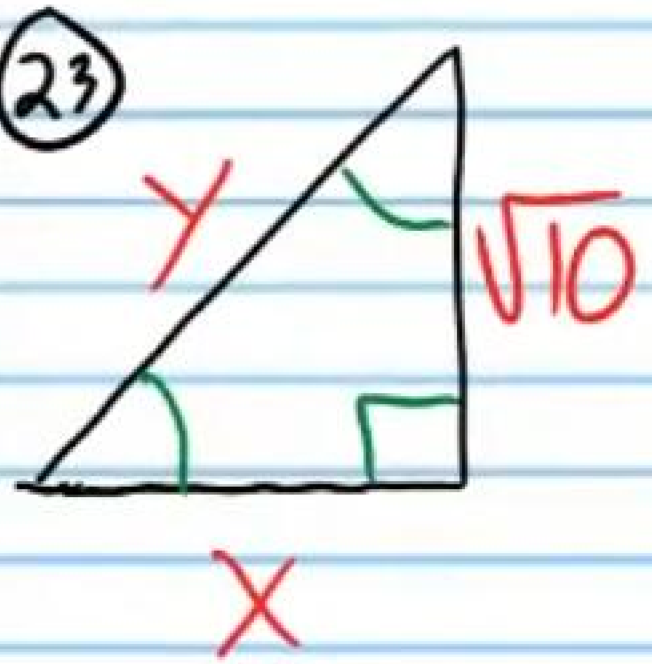
$\boxed{X = 8\sqrt{10}}$

22



$\boxed{X = \frac{15\sqrt{2}}{2}}$

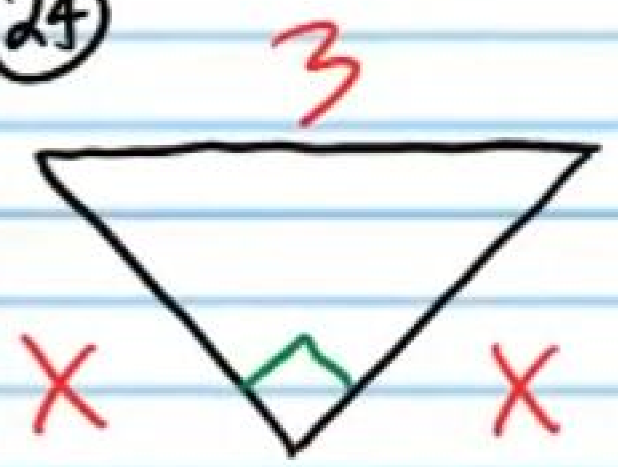
23



$\boxed{X = \sqrt{10}}$

$\boxed{y = 2\sqrt{5}}$

24



$\boxed{X = \frac{3\sqrt{2}}{2}}$